Study Guide, Exercises, and Sample Questions

CSE 4415: Wireless Networks

Islamic University of Technology (IUT)

Lecture 7.1: MAC Efficiency Boost — Frequency-Domain Backoff (Back2F)

1. Topics Covered

- Review of current WiFi channel contention: DIFS wait, random backoff countdown, freeze-and-resume behavior, and per-packet contention.
- Worked contention example: S1 with backoff $B_1 = 15$ and S2 with backoff $B_2 = 25$; countdown, transmission, deferral ("Wait"), resumption from $B_2 = 10$, new draw $B_1 = 18$, and the next round where S2 wins.
- Channel wastage due to temporal backoff: every contention round forces both stations to burn idle slots; approximately 35% overhead at 54 Mbps.
- Quantitative motivation: a 1500-byte frame at 54 Mbps takes $\approx 250\,\mu$s, while the *average* temporal backoff wastes $\approx 100\,\mu$s — a very large relative overhead.
- Key observation: backoff is fundamentally a *randomization* operation, not a time-domain operation; only its *implementation* is in the time domain.
- OFDM in 802.11 a/g/n PHY: wideband channel divided into 48 narrowband subcarriers; motivation is purely PHY (robustness to fast, frequency-selective fading).
- MAC opportunity: treat the 48 OFDM subcarriers as integers $0 \dots 47$ and emulate randomized backoff in the frequency domain.
- Back2F main idea: each station transmits a signal on the subcarrier equal to its random backoff number, while simultaneously using a second (listen) antenna to detect all active subcarriers.
- Winner determination: the station whose active subcarrier has the *lowest* index wins; every station learns the full set of chosen backoffs and hence its own *rank*.
- Frequency backoff cost: one contention round = 1 OFDM symbol = $4\,\mu$s, versus $\approx 100\,\mu$s average temporal backoff.
- Collisions in the frequency domain: two stations choosing the same subcarrier; resolution via a *second round* of contention among first-round winners.
- Optimizing for TDMA: admitting not only the winners but a few more stations into the second round so that a longer collision-free TDMA schedule can be built.
- Scheduled (TDMA-like) transmission: each station knows its rank, so stations transmit back-to-back using the *entire channel bandwidth* — contention happens once per TDMA schedule, not once per packet.
- Back2F advantages: fully distributed, low overhead, highly efficient, no idle time before transmission, quick to stabilize; up to 50% throughput improvement.

- Back2F limitations: requires an extra (listen) antenna, sensitive to channel fluctuation/fading on subcarriers, and potential unfairness toward legacy 802.11 devices in mixed deployments.

2. Focus Areas

- Per-packet contention in 802.11: be able to trace the full S1/S2 timeline (DIFS $\rightarrow$ countdown $\rightarrow$ Data+ACK $\rightarrow$ deferral $\rightarrow$ resumed countdown $\rightarrow$ next winner) and identify *where* idle slots are wasted.
- Overhead arithmetic: frame transmission time $T_{\text{data}} = \frac{L \times 8}{R}$, average temporal backoff $\approx 100\,\mu$s, frequency backoff $= 4\,\mu$s; efficiency $\eta = \frac{T_{\text{data}}}{T_{\text{data}} + T_{\text{overhead}}}$.
- The conceptual separation between *what backoff achieves* (randomized ordering of contenders) and *how it is implemented* (idle time slots vs. subcarrier indices).
- Mapping backoff numbers to subcarriers: random integer in $\{0, \ldots, 47\} \rightarrow$ transmit on that subcarrier for one OFDM symbol; lowest active subcarrier wins.
- Role of the listen antenna: simultaneous transmit-and-listen is what allows every station to observe *all* chosen backoffs in a single symbol — something impossible in time-domain backoff.
- Rank computation: given the set of active subcarriers, a station's rank = 1 + (number of active subcarriers with a strictly smaller index than its own).
- How rank is derived from subcarrier occupancy: lowest active subcarrier index wins; how a station learns *both* its own choice and everyone else's via the listen antenna.
- Why a second round exists, and why admitting "a few more stations" (not just winners) into the second round makes the TDMA schedule longer and more effective.
- Contention per packet (802.11) vs. contention per TDMA schedule (Back2F): amortization of contention cost over k packets.
- TDMA transmission uses the full channel bandwidth — the subcarriers are used only for *signaling* during contention, not for data.
- Advantages and limitations lists: be able to reproduce all five advantages and all three limitations with one-line explanations.
- Hardware and environmental assumptions Back2F depends on (full-duplex-like listen/transmit capability, subcarrier fidelity under fading) and why these are also its main limitations.

3. How to Study These Topics

1. Redraw the complete S1/S2 contention timeline from the lecture ($B_1 = 15$, $B_2 = 25$) from memory, labeling DIFS, every countdown segment, Data+ACK, Wait, the resumed value $B_2 = 10$, the fresh draw $B_1 = 18$, and the second winner.

2. Compute, by hand, the transmission time of a 1500-byte frame at 54 Mbps and compare it against the $\approx 100\,\mu$s average temporal backoff; express the backoff as a percentage of the frame time.

3. Draw the frequency axis $0 \ldots 47$ and place example backoff values on it (e.g., 6 and 18); annotate which station wins and what each listen antenna observes.

4. Practice rank computation: invent five random subcarrier sets (e.g., $\{2, 9, 12, 30, 41\}$) and compute the rank of each station.

5. Derive the efficiency formulas $\eta_{802.11} = \frac{T_{\text{data}}}{T_{\text{data}}+T_{\text{backoff}}}$ and $\eta_{\text{Back2F}} = \frac{k\,T_{\text{data}}}{k\,T_{\text{data}}+T_{\text{freq}}}$ and evaluate them for several values of k (schedule length).

6. Work through the two-round contention example: two stations colliding on subcarrier 1 in round one, then re-drawing in round two; compute the probability that they collide again.

7. Prepare one-page handwritten summaries of:

   - The Back2F contention procedure (draw, transmit on subcarrier, listen, rank, TDMA).
   - The overhead comparison table: 802.11 temporal backoff vs. Back2F frequency backoff.
   - The five advantages and three limitations of Back2F.

4. Exercises

4.1. 802.11 Temporal Backoff Review (with Numericals)

1. Two stations S1 and S2 finish DIFS simultaneously with backoffs $B_1 = 15$ and $B_2 = 25$. (a) Which station transmits first? (b) What is S2's remaining backoff when S1 begins transmitting? (c) After S1's Data+ACK completes and the channel is idle again for DIFS, S1 draws a new backoff of 18 while S2 resumes from its frozen value. Which station wins the second round, and after how many slots?

2. Continuing the previous scenario: when S2 transmits (having counted down from 10 to 0), S1 has counted its own backoff down from 18. (a) What is S1's remaining backoff when S2 starts transmitting? (b) The lecture shows $B_1 = 8$ at that point — verify this by counting slots.

3. A slot lasts $9\,\mu$s (802.11a). Compute the idle time wasted in round one of the above scenario (15 slots) and round two (10 slots), in microseconds. Add the two DIFS periods ($34\,\mu$s each) and report total contention overhead across both rounds.

4. Three stations draw backoffs 4, 4, and 11. (a) What happens after 4 slots? (b) What do the two colliding stations do next, and what does station three do with its remaining count? (c) Why does this failure mode motivate a mechanism in which every station can *see* all chosen backoff values?

5. A 1500-byte frame is sent at 54 Mbps. (a) Compute the raw transmission time $T_{\text{data}} = \frac{1500\times8}{54\times10^6}$ in microseconds. (b) With an average temporal backoff of $100\,\mu$s, compute the

efficiency $\eta = \frac{T_{\text{data}}}{T_{\text{data}}+100\,\mu\text{s}}$. (c) Reconcile your answer with the lecture's claim of roughly 35% total overhead at 54 Mbps (note: total overhead also includes DIFS, SIFS, ACK, and headers).

6. Repeat exercise 5(a)–(b) for a small frame of 200 bytes at 54 Mbps. What happens to efficiency as the frame gets smaller? What does this imply about backoff overhead for VoIP-style traffic?

4.2. From Time Domain to Frequency Domain (Conceptual)

7. The lecture states: "Backoff is not fundamentally a time domain operation — its implementation is in the time domain." Explain this statement. What is the *fundamental* purpose of backoff, and why can that purpose be served in any domain that supports randomization?

8. What "MAC opportunity" does Back2F extract from this purely-PHY design decision?

9. How many subcarriers does the OFDM PHY divide the wideband channel into? How does Back2F reinterpret these subcarriers?

10. Why does time-domain backoff *require* the channel to remain idle during countdown, and why does frequency-domain backoff not waste equivalent idle time?

4.3. Back2F Main Idea and Rank Determination (with Numericals)

11. Describe the two roles a station's radio must play simultaneously in Back2F, and explain why this requires (at least conceptually) two antennas.

12. Two stations, S1 and S2, independently and uniformly pick a subcarrier index from $\{0, 1, \ldots, 47\}$. S1 picks 6; S2 picks 18. State which station is the winner of this round according to the rule described in the lecture, and explain how *both* stations come to know this without any central coordinator.

13. Four stations pick subcarriers $\{31, 4, 22, 4\}$ respectively (stations A, B, C, D). (a) Which station wins outright? (b) Which two stations are tied, and why can this tie *not* be resolved within the same round? (c) What must happen next?

14. Explain why the winner is defined as the station on the *lowest-indexed* active subcarrier rather than, say, the highest, or a subcarrier chosen by some other rule. Does the specific ordering convention matter conceptually?

15. S1 draws backoff 6 and S2 draws backoff 18. Both transmit on their chosen subcarriers in the same OFDM symbol while listening with a second antenna. (a) What does S1's listen antenna observe? (b) What does S2's listen antenna observe? (c) Which station wins, and crucially, does the *loser* also know it lost? Contrast this with time-domain backoff, where a loser merely observes a busy channel.

16. Five APs choose subcarriers $\{3, 7, 12, 29, 44\}$. (a) Determine the rank of the AP that chose 12. (b) The AP that chose 44 — when will it transmit in the resulting TDMA order? (c) If a sixth AP joins and picks subcarrier 0, how does every other AP's rank change?

17. An AP transmits on subcarrier 12 and hears active subcarriers at $\{2, 8, 12, 27, 40\}$ (including its own). The lecture slide labels this "Rank in TDMA: 3." Verify this rank by counting, and state the general rule for computing rank from the observed subcarrier set.

18. One contention round costs 1 OFDM symbol = $4\,\mu$s. (a) With a single round, how much contention time does Back2F spend before a TDMA schedule of 3 stations begins? (b) With two rounds (collision resolution), how much? (c) Compare against three separate 802.11 contentions at $\approx 100\,\mu$s each for the same three packets.

19. Using $T_{\text{data}} = 250\,\mu$s per packet: compute total channel time to deliver 3 packets under (a) 802.11 per-packet contention ($100\,\mu$s before each packet) and (b) Back2F with one $4\,\mu$s contention round before the whole TDMA schedule. Report the throughput improvement as a percentage, and compare with the lecture's "up to 50%" claim.

20. Generalize exercise 15: for k packets of duration T_{data} each, write efficiency expressions $\eta_{802.11}(k)$ and $\eta_{\text{B2F}}(k)$, then evaluate both for $k = 1, 3, 10$ with $T_{\text{data}} = 250\,\mu$s. What does $k \to \infty$ give for each scheme?

4.4. Frequency-Domain Collisions and the Second Round (with Numericals)

21. In the lecture example, two stations (red and green) both choose subcarrier 1 while three others choose subcarriers 3, 4, and 5. (a) Why can neither the red nor the green station detect this collision by itself during round one? (b) How does the second round of contention resolve their tie? (c) What happens if they collide *again* in the second round?

22. Two stations collide on the same subcarrier in round one and independently re-draw uniformly from 48 subcarriers in round two. (a) What is the probability that they collide again in round two? (b) What is the probability of colliding in both rounds, assuming independent uniform draws each round? (c) Compare this with the time-domain collision situation, where a collision is only discovered after a full (failed) data transmission. $P(\text{no collision}) = \prod_{i=0}^{4} \frac{48-i}{48}$. Evaluate numerically. Then repeat for $N = 10$. Compare qualitatively with the lecture-5 claim that 10 time-domain contenders exceed 40% collision probability.

23. Why would a second round with *only* the first-round winners often make "TDMA not effective"? What does the lecture propose instead, and how does admitting a few extra stations improve channel utilization?

24. In the "Optimize for TDMA" example, the stations at subcarriers 1 (two colliding), 3, 4, and 5 participate, and after the second round the ranks are: purple $\to$ Rank 1, red $\to$ Rank 2, green $\to$ Rank 3. Explain how second-round subcarrier choices (shown as small markers 0, 2, 4) determine this final ordering.

25. In the lecture's worked example, stations on subcarriers 1 (two stations tied) and $3, 4, 5$ (single winners) exist after the first round. Draw the corresponding second-round diagram in which only the tied station(s) on subcarrier 1 re-contend. What is the minimum number of stations that must participate in a second round for it to be meaningful?

4.5. Scheduled TDMA Transmission and Channel Utilization

26. Explain the problem statement "Only a Few Stations in Second Round?" from the lecture. Why does the lecture conclude that "TDMA will not be effective" if only the strict tie-breakers advance?

27. In the "optimize for TDMA" scheme, instead of only the round-1 tied winner(s) advancing, the *lowest three* distinct subcarrier holders from round 1 (e.g., ranks corresponding to subcarriers 1, 3, and 4 in the lecture's diagram) advance to round 2, each re-picking from a reduced set $\{0, 2, 4\}$. If their round-2 picks are $\{2, 0, 4\}$ respectively, state the final TDMA order (Rank 1, Rank 2, Rank 3) of the three stations.

28. Suppose 6 stations contend in round 1 and the protocol always advances the lowest-indexed 3 stations to round 2 (regardless of whether they tied). (a) What happens to the 3 stations that were *not* in the lowest three? (b) Propose (in words) how they should be handled in the next transmission opportunity.

29. Discuss the trade-off in choosing how many stations advance per round: what happens to TDMA schedule quality if this number is too small (e.g., always exactly 1)? What happens if it is too large (e.g., all N contenders every round)?

30. Draw the two timelines from the lecture side by side: (a) 802.11 with contention (red slots) before *every* Data/ACK, and (b) Back2F with one short frequency backoff before a train of three Data/ACK bursts. Annotate where channel time is saved.

31. A TDMA schedule serves ranks 1, 2, and 3 back-to-back. Station 2's Data/ACK exchange fails mid-schedule due to channel fading. Discuss: does the failure of one station's slot disrupt the ranks of the remaining stations? What re-contention behavior would be reasonable?

32. Suppose 6 stations contend, each round costs $4\,\mu$s, and 2 rounds are used; the resulting TDMA schedule serves all 6 stations with $250\,\mu$s Data/ACK each. Compute total time, per-packet effective overhead, and efficiency. Compare with 802.11 spending $100\,\mu\text{s} + 250\,\mu\text{s}$ per packet.

4.6. Advantages, Limitations, and Design Trade-offs

33. List the five advantages of Back2F stated in the lecture and give a one-sentence justification for each (fully distributed; low overhead; highly efficient; no idle time before transmission; quick to stabilize).

34. List the three limitations of Back2F. For each, explain the underlying technical cause: (a) why is a second antenna required? (b) why does channel fluctuation threaten subcarrier-based signaling? (c) why might legacy 802.11 devices be treated unfairly in a mixed deployment?

35. A deep fade wipes out the frequency band containing subcarriers 0–5 at a listening station, while another station signals its backoff on subcarrier 2. What incorrect conclusion might the listener draw about its own rank, and what MAC-level failure can result?

36. Compare Back2F's collision-free TDMA schedule with 802.11 PCF (from Lecture 5). Both produce contention-free periods — yet Back2F is called "fully distributed" while PCF is not. Explain the difference in coordination models.

37. Legacy 802.11 stations perform CCA and defer while Back2F stations exchange subcarrier symbols and then launch a long TDMA train. Explain step by step why the legacy stations may starve, and propose one coexistence mechanism.

5. Sample Questions

5.1. Short Questions

1. What is the approximate MAC overhead of 802.11 at 54 Mbps, and what is its dominant cause according to this lecture?
2. How long does a 1500-byte frame take at 54 Mbps, and how long is the average temporal backoff?
3. How many OFDM subcarriers does the 802.11 a/g/n PHY provide, and how are they indexed in Back2F?
4. What was the original (PHY) motivation for adopting OFDM in 802.11 a/g/n?
5. In one sentence, state the main idea of Back2F.
6. What is the duration of one frequency-domain backoff round, and what does it correspond to physically?
7. What hardware element allows a Back2F station to learn the backoff values of all other contenders?
8. Given active subcarriers $\{5, 11, 23\}$, which station wins and what is the rank of the station on subcarrier 23?
9. How does Back2F resolve two stations choosing the same subcarrier?
10. Why are "a few more stations" (beyond the winners) admitted into the second round?
11. After ranks are fixed, over what bandwidth does each station transmit its data?
12. How often does 802.11 contend for the channel, and how often does Back2F contend?
13. State the throughput improvement claimed for Back2F.
14. Name the three limitations of Back2F.

5.2. Descriptive Questions

15. Describe the complete 802.11 contention cycle between two stations (backoffs 15 and 25) across two rounds, including freeze-and-resume, and identify every interval in which the channel sits idle. Then explain, quantitatively, why the lecture calls this "high channel wastage due to backoff."

16. Explain the complete Back2F contention procedure: random draw, subcarrier signaling, simultaneous listening, winner determination, rank computation, and the transition into TDMA transmission. Illustrate with backoffs 6 and 18.

17. Derive and compare the efficiency of 802.11 (100 μs contention per 250 μs packet) and Back2F (4 μs contention per TDMA schedule of k packets). Show how the advantage grows with k and connect it to the two timeline diagrams ("contention per packet" vs. "contention per TDMA schedule").

18. Discuss all five advantages and all three limitations of Back2F. For each limitation, suggest one mitigation and comment on its cost.

5.3. Analysis and Design Questions

19. Four APs contend using Back2F and draw subcarriers 9, 21, 9, and 36. (a) Identify the round-one collision. (b) The two colliding APs re-draw in round two, obtaining 14 and 3; the other APs (21 and 36) also join the second round. Determine the final TDMA order. (c) Compute total contention time (2 rounds $\times$ 4 μs) and the total time to serve all four with 250 μs Data/ACK each. (d) Compute the same workload under 802.11 assuming 100 μs average backoff per packet and no collisions, then report the percentage improvement.

20. In a dense deployment, $N = 20$ stations contend with $K = 48$ subcarriers. (a) Compute $P(\text{all distinct}) = \prod_{i=0}^{19} \frac{48-i}{48}$ (a rough numerical estimate suffices). (b) Explain why frequency-domain collisions are far cheaper than time-domain collisions even when they occur. (c) Propose how the number of contention rounds should scale with N, and what the cost of each extra round is.

21. A Back2F network serves k packets per TDMA schedule. Channel coherence time limits a schedule to 2 ms. With $T_{\text{data}} = 250\,\mu$s per packet and 2 contention rounds of 4 μs: (a) find the maximum k per schedule; (b) compute efficiency at that k; (c) discuss what happens to the schedule (and to the "Sensitive to Channel Fluctuation" limitation) if coherence time drops to 0.5 ms.

22. Design question: Back2F requires simultaneous transmit and listen. A vendor proposes instead to *alternate*: transmit the subcarrier in symbol 1, listen in symbol 2. (a) Does this preserve the ability to compute rank? (b) What new failure appears if all stations transmit in symbol 1 and listen in symbol 2? (c) Propose a randomized fix and estimate its added overhead in OFDM symbols.

23. Evaluate the end-to-end trade-off between 802.11 DCF and Back2F. Construct a table comparing the two across: contention overhead per packet, collision detection latency, hardware requirements, robustness to fading, fairness to legacy devices, and behavior at high node density. Under what deployment conditions would you still prefer plain DCF?

24. Six stations independently pick subcarriers as follows: A= 14, B= 2, C= 39, D= 2, E= 27, F= 9. (a) Determine the round-1 winner. (b) Identify any tie(s) and state which station(s) cannot yet be placed in the final order. (c) If the protocol advances the three lowest-indexed *distinct* subcarrier values (with ties counted as a single group needing resolution) to round 2, list which stations advance, and design a round-2

subcarrier set they should choose from so that the tie between D and B can be broken.

25. Ten stations each have one 250 μs packet ready. Compare total channel-occupied time for: (a) legacy DCF, where each station separately incurs a 100 μs average backoff before its own packet; (b) Back2F with a *single* contention round (one 4 μs frequency backoff/ranking phase, all ties assumed resolved) followed by all ten packets sent back-to-back with no further contention. Compute the total time for each scheme and the percentage improvement Back2F achieves.

26. Let $N = 4+(X \bmod 4)$ be the number of contending stations and assume 48 available subcarriers. (a) Using digits X and Y, construct N distinct subcarrier picks for your stations (e.g., using a simple formula such as subcarrier$_i$ $= (X \cdot i + Y) \bmod 48$ for $i = 1, \ldots, N$, adjusting any accidental duplicate by $+1$). List the resulting subcarrier assignments. (b) Determine the round-1 winner. (c) State whether any ties exist under your assignment; if not, briefly modify one station's pick to deliberately create a tie, and describe how a second round would resolve it. (d) Using 250 μs packets and a 4 μs frequency backoff, compute the total channel time for your N stations under a single-round Back2F TDMA schedule, and compare it to the legacy DCF estimate (using 100 μs average temporal backoff per station).